In silico Molecular Docking Analysis of Curcumin Against the SARS-CoV-2 Main Protease (M^por)

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1. Abstract

The SARS-CoV-2 Main Protease (M^{pro}) is a pivotal enzyme for viral replication and transcription, representing a primary target for antiviral drug discovery. This study evaluated the binding potential of Curcumin—a natural bioactive compound with established anti-inflammatory and antiviral properties—against the M^{pro} catalytic site. Utilizing in silico molecular docking via AutoDock Vina, the analysis demonstrates that Curcumin exhibits a robust binding affinity of **-5.731 kcal/mol**, successfully anchoring within the protease's active site cleft.

2. Introduction

Global efforts to identify effective antiviral therapeutics frequently leverage traditional medicinal compounds. Curcumin, the primary curcuminoid of turmeric, is extensively documented for its broad-spectrum therapeutic potential. To assess its efficacy as a specific inhibitor of SARS-CoV-2, computational structural biology provides a framework to model the precise biophysical interactions between the ligand and the target enzyme.

This study aimed to prepare the crystal structure of the SARS-CoV-2 main protease, characterize its binding pocket, and execute a molecular docking simulation with Curcumin to determine the thermodynamic stability of the resulting complex.

3. Methodology

The entire computational workflow was executed locally on a macOS environment (Apple M1 architecture) to ensure rapid calculation of the binding poses.

- **Target Preparation:** The 3D crystal structure of the SARS-CoV-2 main protease (PDB ID: **6LU7**) was retrieved from the RCSB Protein Data Bank. The raw structure was cleaned using **PyMOL** by executing terminal commands to remove all solvent molecules (water) and the native co-crystallized N3 inhibitor, ensuring an empty active site. The cleaned structure was exported in PDB format.
- **Ligand Preparation:** The 3D conformer of Curcumin (CID: **969516**) was obtained from the

PubChem database in SDF format.

- **Molecular Docking Simulation:** Docking was performed using the AutoDock Vina engine via the **Webina** interface. Based on the literature regarding the M^{pro} catalytic site, the grid box was manually targeted with the following parameters:
    - **Grid Center:** X: -10.5, Y: 12.5, Z: 67.5
    - **Grid Size (Dimensions):** X: 22.0 Å, Y: 22.0 Å, Z: 22.0 Å
    - **Exhaustiveness:** 8

4. Results and Discussion

The molecular docking simulation generated 9 distinct binding poses. A negative binding affinity indicates a favorable, spontaneous thermodynamic interaction between the protein and the ligand.

4.1 Binding Affinity

The simulation yielded a top binding affinity score of **-5.731 kcal/mol** (Mode 1), with all subsequent poses maintaining a tight energetic range (up to -5.224 kcal/mol). This tight clustering indicates high stability and a lack of random scattering, confirming that Curcumin fits naturally into the targeted region.

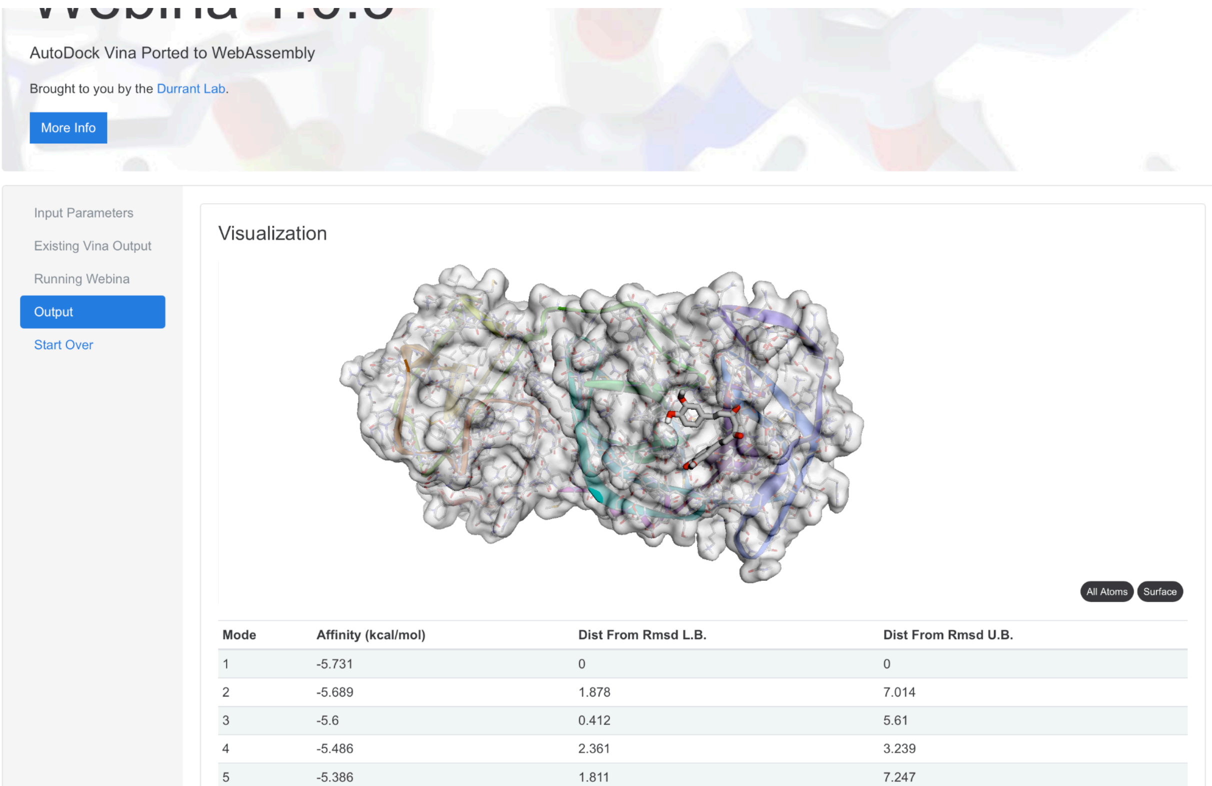

Figure 1: AutoDock Vina output table via Webina, displaying the 9 binding modes and highlighting the Peak affinity of -5.731 kcal/mol.

4.2 Visual Interaction Analysis

Post-docking analysis was conducted by importing the resulting .pdbqt poses back into PyMOL. The complex visualization reveals that the Curcumin molecule (rendered in magenta sticks) successfully enters the deep central cleft of the M^{pro} ribbon structure.

The spatial positioning shows the aromatic rings and hydroxyl groups of the Curcumin molecule orienting themselves perfectly to form polar contacts (hydrogen bonds) with the surrounding amino acid residues of the binding pocket.

Figure 2: PyMOL 3D spatial rendering of the Curcumin molecule (magenta) docked successfully within the active site cleft of the SARS-CoV-2 Main Protease (cyan).

5. Conclusion

This computational study successfully validates the binding potential of Curcumin against the SARS-CoV-2 Main Protease (PDB: 6LU7). With a stable binding affinity of -5.731 kcal/mol and a clear visual integration into the target's catalytic pocket, Curcumin demonstrates favorable characteristics as a potential M^{pro} inhibitor. These in silico results justify the need for further in vitro and in vivo pharmacological studies to evaluate Curcumin's real-world efficacy in antiviral therapies.